

Sergei Kushneriuk

Latest version of my CV can always be found [here](#)

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Education

2023 – 2025 **Master's Degree in Data Science with Distinction.**

Joint "Math of Machine Learning" program at [Skoltech](#) and [HSE University](#), GPA: 4.92/5.0

- Markov Chains, High Dimensional Probability and Statistics, Optimization methods for ML
- Machine Learning and Deep Learning, Generative Models, Bayesian Methods of Machine Learning, Models of Sequential Data, Geometrical Methods of Machine Learning

2021 – 2025 **Machine learning developer academic program, [Yandex School of Data Analysis](#).**

- Advanced courses in Classical ML and DL, RL, CV and NLP, Bayesian Statistics and ML

2019 – 2023 **Bachelor's Degree in Applied Mathematics and Computer Science.**

[Moscow Institute of Physics and Technology](#), GPA: 4.7/5.0

- Further and Discrete Maths disciplines, Advanced Computer Science and Data Science disciplines
- Student of Department of Image Recognition and Text Processing founded by [ABBY](#): DL, CV, NLP
- Student of Advanced Data Science track: ML, DL, NLP, CV, Advanced Applied Statistics, ANOVA and AB-testing, Bayesian Statistics, Generative Modelling, Time Series and Stochastic Processes, Clustering algorithms and Unsupervised Learning

Working experience

2025 - now **Generative Modelling Researcher, [Huawei Noah's Ark](#), [Moscow AI Foundation](#) and [Algorithm Lab](#), [LLM Visual Team](#), Python, Pytorch, Transformers, vLLM, trl, Ascend NPU.**

- Creating Pipelines for Data Generation for VLM Training: Referring Expression Comprehension and Generation Datasets with VLMs
- Investigation of Discrete Diffusion application to generation of Visual and Text data
- Acceleration and Distillation of Diffusion-based models for Blind Image Super Resolution

summer 2024 **NLP Intern, [Tinkoff AI Center](#), [Applied NLP Team](#), Python, Pytorch, Transformers.**

- Accelerated high-performance NER- and rule-based text anonymization tool
- Extended tools of text clustering service; developed semi-supervised module for text clustering

summer 2022 **ML Intern, [Meteum.AI](#) - [Weather Prediction Team](#), Python, TensorFlow, SQL.**

- Improved short-term precipitation prediction model with (new at the time) users' reports
- Analyzed weather forecasting impact on RecSys algorithms in e-commerce

summer 2021 **SWE Intern, [Yandex](#) - [Weather Back-end Team](#), C++, Python, Go, Google Protobuf.**

- Developed 4 new weather scenarios for voice assistant Alice and set up their AB-testing

2020 – 2021 **Competitive Programming Tutor, [SPGuide](#).**

- Taught individually 10 school-students from 6 to 11 grade basic Algorithms & Data structures, C++, Python

Programming skills and experience

ML/DS Experience at implementing and training different ML/DL pipelines: Classical ML, Clustering, Models for Time Series, Bayesian ML, Transformers, Diffusion models

Python pytorch, transformers, diffusers, catboost/xgboost/lightgbm, other standard DS libraries; FastAPI

SWE Familiar with support and development of High Load Distributed Services

Other C++, Go, Linux & Bash & Git, Docker, $\LaTeX$

Research

2025 **[One-Step Residual Shifting Diffusion for Image Super-Resolution via Distillation \(ICML 2026 Submission\)](#), [Daniil Selikhanovych](#), [David Li](#), [Aleksei Leonov](#), [Nikita Gushchin](#), [Sergei Kushneriuk](#), [Alexander Filippov](#), [Evgeny Burnaev](#), [Iaroslav Koshelev](#), [Alexander Korotin](#), In: arXiv preprint [arXiv:2503.13358](#) (2025).**

Overall: This Paper introduces new method of Distillation of Diffusion Model for BlindSR (teacher) into One-Step Generator (student). Main idea is to train student model in such a way, so that additional "fake" model trained on student's output data will coincide with teacher model.

Personal contribution:

- Reproduced and compared existing Diffusion Models Distillation approaches and other One-Step competitors
- Conducted experiments on GAN-losses, their impact on training and appropriate hyperparameter selection for them

- 2025 **Improving Diffusion-based Blind Image Super-Resolution Methods via Invertible Image Transformations**, Graduate qualifying Master's Thesis, under supervision of [Alexander Korotin](#).
- Combined ideas of Diffusion Bridge Matching Models and classic Discrete Cosine and Wavelet Transforms to improve performance on the BlindSR task.
 - Introduced frequency-aware DWT/DCT-based losses, DWT/DCT-space Diffusion Bridge Process and Blurring Bridge Process
- 2023 **Comparative analysis of techniques for improving the performance of Semantic Segmentation models under class imbalance conditions**, Graduate qualifying Bachelor's Thesis.
- Reproduced results of the paper [Unified Focal Loss](#) under new conditions: new (non-medical) datasets and class imbalance levels
 - Compared performance of wide range of loss functions
 - Identified when Unified Focal Loss doesn't give any advantages

Achievements

- 2024 **ICPC Semifinals - Northern Eurasia Finals**, 2nd degree diploma, place 53/269.
- 2023 **ICPC Semifinals - Northern Eurasia Finals**, 3rd degree diploma, place 74/276.
- 2023 **All-Russian university student Olympiad "I Am a Professional"**, Maths - 1st degree, AI - 2nd degree.
- 2023 **All-Russian university student Olympiad "The Higher League"**, Applied Maths and Computer science - 2nd degree.

Languages

English C1-C2 Upper-Intermediate
Russian Native Speaker